11. Sequences and Series

11.8. Power Series

11.8 Learning Objectives
- Identify power series
- Use previous techniques to determine whether the series converges or diverges
- Understand how the center and coefficients affect the convergence of power series
- Identify the radius of convergence for series that converge on an interval

11.8.1. Motivation

Do the following series converge or diverge? How do you know?

1. $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$
   converges because it is geometric with $a = 1$ and $|r| = \frac{2}{3} < 1$

2. $\sum_{n=0}^{\infty} (-4)^n$
   diverges because it is geometric with $a = 1$ and $|r| = 4 > 1$

3. $\sum_{n=0}^{\infty} 44\left(\frac{1}{100}\right)^n$
   converges because it is geometric with $a = 44$ and $|r| = \frac{1}{100} < 1$

Observe that r is changing in these scenarios, so we can represent these with a variable x...

11.8.2. Power Series

Definition 11.8.1 Power Series

A **power series** is a series of the form:

$$\sum_{n=0}^{\infty} c_n x^n$$

where x is a variable, and c_n are constants called the **coefficients** of the series.

We can also write power series as $\sum_{n=0}^{\infty} c_n (x-a)^n$ where a is a constant we call the center.

Note

- Notice a does not have a dependency on n, so it is the same for every term in the series.
- Here c_n depends on which term in the series, n we are currently adding, so c_0 may be different from c_1 etc.
- You could write this as

  $$\sum_{n=1}^{\infty} c_n x^{n-1}$$

  if you like, but we won't. We will just start these at 0.
- Power series is named because the terms are all some number times a power of x.

- Examples of power series
  - $\sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \ldots$ is a power series where the first coefficient $c_0 = 0$, the center $a = 0$, and $c_n = \frac{1}{n}$ for $n \geq 1$.
  - $\sum_{n=0}^{\infty} 3(x-1)^n = 3 + 3(x-1) + 3(x-1)^2 + \ldots$ is a power series where $c_n = 3$ for all $n \geq 0$ and center $a = 1$. (The sum started at 0 this time, like we expected, so we did not have to write the first coefficient was 0 like the last example)
  - $\sum_{n=0}^{\infty} (n+1)!x^n = 1 + 2x + 6x^2 + 24x^3 + \ldots$ is a power series where $c_n = (n+1)!$ and center $a = 0$.
- What affects the convergence or divergence of this series?
  - The value of x and the values of the sequence $(c_n)_{n=1}^{\infty}$ impacts the convergence or divergence. x is just a number, but if c_n depends on n and is not just a number, then the series cannot be considered like geometric.
- If $x = 0$, does the series $\sum_{n=0}^{\infty} c_n x^n$ converge?
  - If $x = 0$, we have $\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 \cdot 0 + 0 + 0 + 0 + 0 + \ldots = c_0$ so the series always converges when $x = 0$.
- Similarly, if we have the series $\sum_{n=0}^{\infty} c_n (x-a)^n$, it always converges for $x = a$. This is why a is called the center of the series.

Definition 11.8.2 **Power Series as a Function**

A power series is like a function

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$$

whose domain is all values of x where the series converges.

Note

For

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n,$$

$x = a$ is always in the domain of $f(x)$.

Example 11.8.1

Find the values of x for which the series $\sum_{n=0}^{\infty} (-1)^n x^n$ converges.

One effective way to find the values of x where a power series converges is with the Ratio Test. This will be a common approach when determining when (what values of x) power series converge.

Solution

By ratio test, if we want the series to converge we would have

$$L = \lim_{n\to\infty}\left|\frac{(-1)^{n+1}x^{n+1}}{(-1)^n x^n}\right| = \lim_{n\to\infty}|x| = |x|$$

Ratio test says that when $L < 1$ we have convergence, $L > 1$ divergence, and it is inconclusive when $L = 1$. So we can start to build up when this series converges. When $L = |x| > 1$ it diverges, when $L = |x| < 1$ it converges, and when $L = |x| = 1$ we have to use a different test.

$|x| = 1$ means $x = -1, 1$. So, checking when $x = 1$ the series is equivalent to $\sum_{n=0}^{\infty}(-1)^n$ which diverges by the Divergence Test since $\lim_{n\to\infty}(-1)^n$ diverges.

Checking when $x = -1$, the series is equivalent to

$$\sum_{n=0}^{\infty}(-1)^n(-1)^n = \sum_{n=0}^{\infty}(-1)^{2n} = \sum_{n=0}^{\infty}1^n = \sum_{n=0}^{\infty}1$$

by exponent rules. Also by the divergence test, $\lim_{n\to\infty}1 = 1 \neq 0$, so the series diverges when $x = -1$.

Therefore, the power series converges exactly when x is in $(-1, 1)$.

In this example, $(-1, 1)$ is called the **interval of convergence** of the power series.

Definition 11.8.3 **Radius of Convergence and Interval of Convergence**

For a given power series $\sum_{n=0}^{\infty}c_n(x-a)^n$ *exactly one* of the following statements is true:

1. The series only converges when $x = a$. Then $R = 0$ is the radius of convergence. The interval of convergence is just the single point $x = a$.

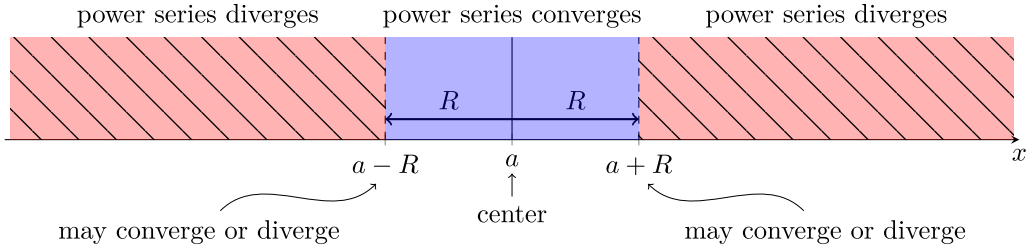
2. The series converges for all real numbers x. Then $R = \infty$ is the radius of convergence and $(-\infty, \infty)$ is the interval of convergence.

3. There is a positive real number R where the series converges for $-R + a < x < a + R$ and diverges for $x > a + R$ and $x < a - R$.

In all cases, the constant R is called the **radius of convergence** for the series $\sum_{n=0}^{\infty}c_n(x-a)^n$, and the **interval of convergence** for the series is the interval consisting of all values of x for which the series converges.

In case 3, when $R > 0$ is finite, a power series could converge or diverge at either endpoint of the interval of convergence. So the interval of convergence for the series is one of the following:

$$(a-R, a+R), [a-R, a+R), (a-R, a+R], \text{ or } [a-R, a+R]$$



Example 11.8.2

Determine the radius of convergence and the interval of convergence for the series

$$\sum_{n=0}^{\infty} \frac{(2x-3)^n}{n!}$$

Solution

We will start with the ratio test,

$$L = \lim_{n\to\infty} \left| \frac{(2x-3)^{n+1}}{(n+1)!} \cdot \frac{n!}{(2x-3)^n} \right| = \lim_{n\to\infty} \frac{|2x-3|}{n+1}$$

For any value of x, the limit as $n \to \infty$ makes this limit $L = 0$. Therefore, this series converges for all x. So, the radius of convergence is $r = \infty$ and the interval of convergence is $(-\infty, \infty)$.

Note

Say in the above example, that the radius was not $R = \infty$, but $R = 5$. We would then need to find the center of the power series in order to determine the interval of convergence. Because of the $(2x-3)$ it is a little trickier, but we set this equal to 0 and solve for a:

$$2a - 3 = 0 \Rightarrow 2a = 3 \Rightarrow a = \frac{3}{2}$$

The center would be $\frac{3}{2}$, and the interval would start as $\left(\frac{3}{2} - 5, \frac{3}{2} + 5\right)$. Then we would have to test each end point separately.

We will see this process in the next example.

Example 11.8.3

Determine the radius of convergence and the interval of convergence for the series

$$f(x) = \sum_{n=0}^{\infty} \frac{(x-1)^n}{5^n(n+1)}$$

Solution

We will once again start with the ratio test,

$$\begin{aligned} L &= \lim_{n\to\infty} \left| \frac{(x-1)^{n+1}}{5^{n+1}((n+1)+1)} \cdot \frac{5^n(n+1)}{(x-1)^n} \right| \\ &= \lim_{n\to\infty} |x-1| \frac{n+1}{5(n+2)} \\ &= \frac{1}{5}|x-1| \end{aligned}$$

This converges when $L < 1$, or when

$$\frac{1}{5}|x-1| < 1$$
$$\Rightarrow |x-1| < 5$$

Thus, the radius of convergence is $R = 5$. Since this power series is centered at $x = 1$, then the end points of the interval are $x = -4,$ and $x = 6$. We need to check these values separately.

When $x = -4$, the series is

$$f(-4) = \sum_{n=0}^{\infty} \frac{(-4-1)^n}{5^n(n+1)} = \sum_{n=0}^{\infty} \frac{(-1)^n(5)^n}{5^n(n+1)}$$
$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)}$$

Because this is an alternating series, we can apply the alternating series test. Notice $\frac{1}{n+1}$ is decreasing and $\lim_{n\to\infty} \frac{1}{n+1} = 0$, so by the alternating series test, $\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)}$ converges, so -4 is in the domain of f.

When $x = 6$, the series is

$$f(6) = \sum_{n=0}^{\infty} \frac{(6-1)^n}{5^n(n+1)} = \sum_{n=0}^{\infty} \frac{(5)^n}{5^n(n+1)}$$
$$= \sum_{n=0}^{\infty} \frac{1}{(n+1)}$$

Let $b_n = \frac{1}{n}$ and $a_n = \frac{1}{n+1}$, both have positive terms for $n > 1$, then

$$\lim_{n\to\infty} \frac{b_n}{a_n} = \lim_{n\to\infty} \frac{\frac{1}{n}}{\frac{1}{n+1}}$$
$$= \lim_{n\to\infty} \frac{n+1}{n} = 1 > 0 \checkmark$$

By limit comparison test, since $\sum a_n$ diverges (harmonic series), so too does $\sum \frac{1}{n+1}$. Then, $x = 6$ is not in the domain of f.

Finally, putting everything together, the interval of convergence is $[-4, 6)$.

Note

The general process for doing this is

1. Apply a series convergence test to find when the power series converges, almost always ratio test.
2. Get the inequality down to $|x - a| < R$, potentially dividing by any constant in front of x to do so. Then this is the radius of convergence R.
3. Find left and right endpoints $a - R$, and $a + R$.
4. Plug in the left and right endpoints separately, and use other series convergence tests to figure out if the resulting series converges or diverges.
5. Write the interval of convergence, including the endpoint if the respective series converged or excluding it if it diverged.

11.8 Section Summary:

- We defined what a power series was.
- We used ratio test and other series convergence tests to determine the interval of convergence (x values when the series converges) and radius of convergence.